

Do bond markets misprice convexity under stress?

Evidence from EM sovereign bonds during the COVID-19 credit shock

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Fixed income

EM bonds

Data science

118

EM sovereign bonds

5

Years of daily data

11

Countries analyzed

BACKGROUND

Convexity measures how a bond's price responds to yield changes. The standard formula calculates it from promised cash flows.

For investment-grade bonds this works well. For distressed credit, promised and expected cash flows diverge sharply as default risk rises.

Whether this shows up in actual market pricing is an open empirical question.

RESEARCH QUESTION

Does the closed-form convexity formula overstate true price-yield curvature for distressed EM sovereign bonds?

The COVID-19 shock of 2020 created a natural experiment: yields on riskier EM bonds spiked in weeks, producing a wide cross-section of credit stress within the same panel.

DATA AND PIPELINE

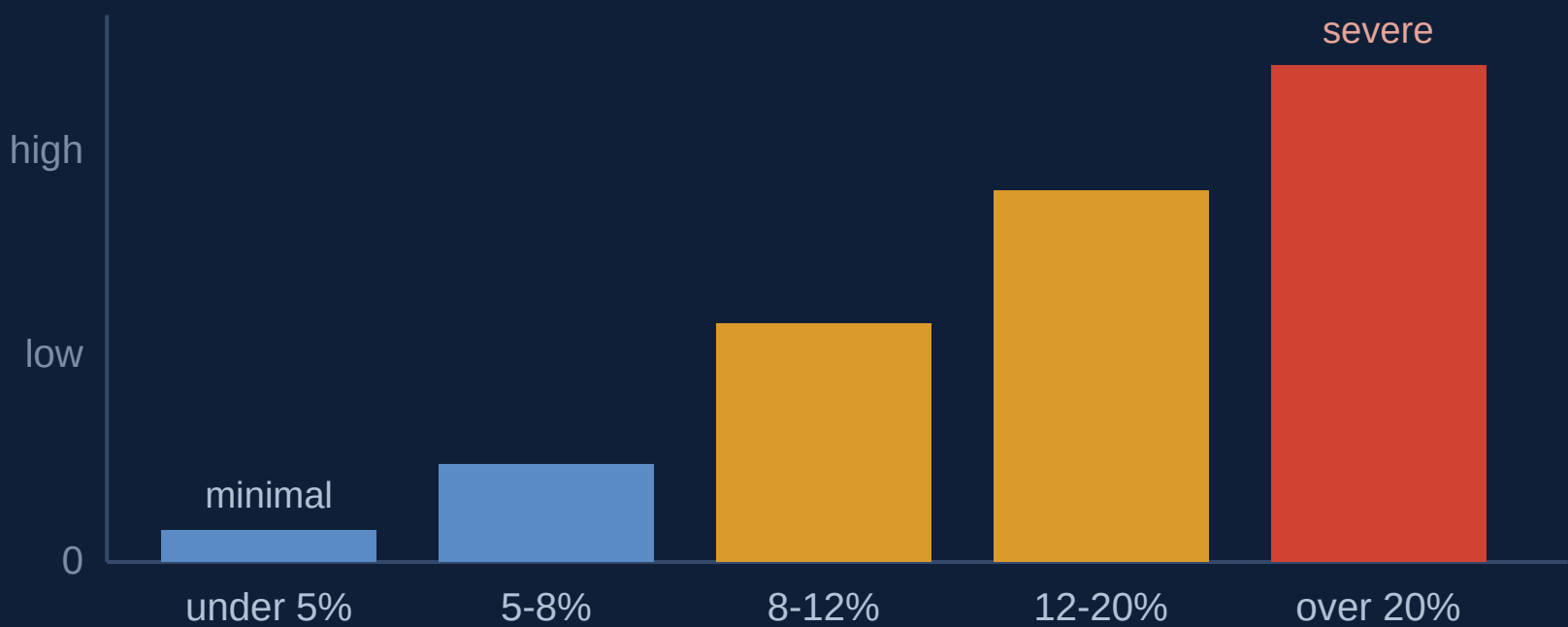
- Refinitiv daily panel, 115 bonds
- Bloomberg data for 3 Argentine bonds post-restructuring
- Convexity scale harmonized across vendor exports
- Formula analytics recomputed from cash flows on locked 30/360 convention
- Restructuring cutoffs applied for Ecuador and Argentina
- Vendor quality flags removed corrupt rows

METHOD AND RESULTS

Formula convexity is computed from cash flows and compared against level-fit curvature estimated from the actual price-yield scatter for each bond-regime.

Three regimes: pre-COVID (before Feb 2020), crisis (Feb to Jun 2020), post-crisis. Bonds grouped into yield buckets to track overstatement across the credit spectrum.

Convexity overstatement by yield bucket



KEY FINDINGS

The formula systematically overstates convexity. The gap widens monotonically as credit risk rises.

At extreme distressed yields, some bonds show locally negative curvature, consistent with the cash-flow argument but not the general rule.

For investment-grade bonds formula and empirical convexity align closely. The divergence is specific to stress.

Result holds across all estimators and yield-bucket specifications.

WHY THE FORMULA OVERSTATES

The formula discounts promised cash flows. A default-risky bond's price reflects expected cash flows, each weighted by survival probability.

As yields rise, survival probability falls. The price-yield curve flattens relative to the formula prediction. True curvature is lower than implied.

IMPLICATIONS

- Hedging ratios based on formula convexity will be overstated for distressed credit
- Risk models using vendor convexity fields uncritically may underestimate tail exposure
- The effect is mild for IG bonds and severe above roughly 8% yield

CONCLUSION

Convexity overstatement is not a modeling artifact but a structural feature of distressed EM sovereign pricing. The finding is robust to convention choice and data source.

Treat formula convexity as an upper bound for bonds trading above 8% yield.